

Emot Act AR: Tailoring Content through User Emotion and Activity Analysis

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ABSTRACT

In an era characterized by immersive content delivery experiences, the convergence of augmented reality (AR) and user behavior data stands out as a transformative synergy. This paper introduces "Emot Act AR," an innovative system intricately designed to dynamically tailor virtual content by user data. This framework incorporates user emotions and activities as intrinsic components. Through the utilization of sensing devices like camera and advanced AI models, the system dynamically customizes virtual elements, such as virtual flowers, in both static and dynamic settings. Additionally, the system utilizes user activity data to activate messaging avatars that encourage exercise.

Keywords: Virtual content generation, Augmented reality, User behavior data analysis, User emotion detection, User activity recognition.

1 INTRODUCTION

In the contemporary landscape, the delivery of content enriched with visualization has emerged as a prominent trend. Augmented Reality (AR), at the forefront of emerging technologies, plays a crucial role in elevating visual experiences by seamlessly integrating computer-generated content into the real world, both directly and indirectly, without delay. This dynamic interaction enables users to actively engage with virtual content [1]. The infusion of AI methods to capture user behavior data further intensifies the synergy between humans and technology [2]. User behavior data encompasses emotions and models situations by discerning activities such as walking or expressions like sadness. This data becomes instrumental in delivering pertinent content and enhancing automated services. Various sensing devices, including wearable sensors like GPS, accelerometers, gyroscopes, cameras, and more, can capture this valuable data [3]. The amalgamation of user behavior data with AR applications serves as a catalyst for the generation of AI-driven content, resulting in a more personalized experience for users. This transformative potential finds practical applications in smart city services. For instance, an application might intricately model the relation between emotion and virtual environments by dynamically altering the color of virtual elements, such as trees and grass, based on the user's emotional state [4]. Another compelling example involves the utilization of head-mounted devices (AR-HMDs) worn by users, empowering them to choose contexts, identify actions, and receive AR content. For instance, the system intelligently detects a user picking up a pill, counts the number of pills, and prompts them to clear the table when they stand up, showcasing the diverse applications of this technology [5].

Acknowledging this convergence, we introduce our groundbreaking system, Emot Act AR. This innovative framework leverages the sensing device like the camera to dynamically alter the content of virtual elements, such as flowers, in both static and dynamic settings. Additionally, the system features a messaging avatar designed to encourage exercise—a concept that will be explored in more detail in the following sections.

2 PROPOSED SYSTEM

Our cutting-edge framework, Emot Act AR, seamlessly integrates sensory input to craft personalized augmented reality (AR) experiences based on user behavior, encompassing both emotions and activities. In the emotion detection segment, our core model utilizes sources such as webcams and employs a sophisticated emotion detection algorithm. This algorithm harnesses the power of OpenCV, Deepface, and Matplotlib modules in Python, along with the MobileNet architecture. It is trained on a comprehensive dataset covering six fundamental emotions (anger, fear, neutral, sadness, disgust, happiness, and surprise). The resultant service dynamically triggers virtual content, such as flowers, tailored to the recognized emotions in both dynamic and static modes. Shifting the focus to activity recognition, our framework captures data from sources like webcams, it then applies an advanced activity recognition model based on Long-Term Recurrent Convolutional Networks (LRCN), utilizing a rich training dataset to identify activities such as sitting standing, and other predefined classes. This aspect of the service activates messaging avatars, encouraging users to participate in various exercise activities. The subsequent section offers an in-depth exploration of the diverse range of services we provide, while Figure 1 visually depicts the components of our innovative framework.

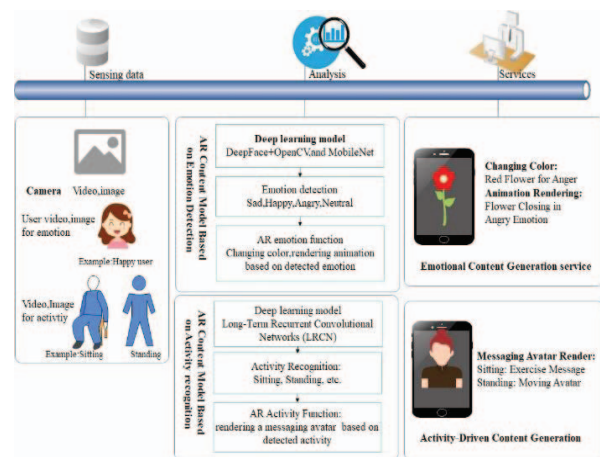


Figure1. Emot Act AR component

2.1. EMOTIONAL CONTENT GENERATION IN AUGMENTED REALITY:

Our innovative service revolutionizes AR content by dynamically transforming virtual elements, such as flowers or animated floral displays, based on the user's emotions, to positively influence and

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uplift their mood. Consider this: When a user expresses anger, the virtual flower turns red; for moments of happiness, it radiates orange; during instances of sadness, it adopts a blue color, and for neutrality, it elegantly presents itself in white within the real space. Going beyond static transformations, our dynamic model introduces animated flower sequences that respond to the user's emotional fluctuations. For example, the virtual flower gracefully blossoms when the user is in a positive mood, such as happiness, and gently closes when they experience negative emotions like anger or sadness.

Figure 2 visually captures outcomes observed in a real case study. In this testing scenario, pre-processed videos are seamlessly integrated into an AR app, providing users with a user-friendly list box interface to dynamically adjust their mood and engage with the emotional content.

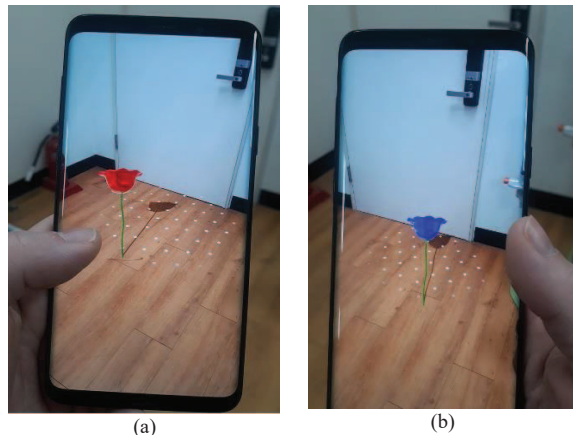


Figure 2. Expected result, a :angry flower, and b :sad flower.

2.2. ACTIVITY-DRIVEN CONTENT GENERATION WITH EMOACT BUDDY:

EmoAct Buddy is designed to inspire users to embrace physical activity. This innovative service monitors the user's activity, utilizing advanced recognition systems. When the system identifies a sitting posture, EmoAct Buddy emerges on the AR app, delivering encouraging messages to prompt the user to explore exercise routines. In contrast, if the recognition system detects a standing posture, EmoAct Buddy springs into action, showcasing the moving avatar to stimulate movement and engagement.

The uniqueness of this service lies in its adaptability to the user's posture, creating a personalized and interactive experience. EmoAct Buddy generates messages and the avatar, enhancing the overall engagement and effectiveness of the exercise encouragement. Figure 3 illustrates the outcomes of these services, featuring an Avatar displaying an exercise message in a sitting position, as well as an Avatar demonstrating exercise movements while standing.

The application of this framework involves utilizing AR SDKs, such as Google Sceneform SDK, along with libraries dedicated to emotion and activity analysis.

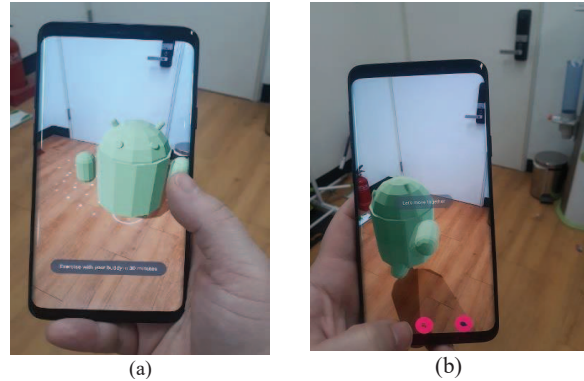


Figure 3. Expected result, a: Avatar with exercise message in a sitting position, and b: Avatar exhibiting exercise movements while standing.

3 CONCLUSION AND DISCUSSION

The Emot Act AR framework represents an innovative application that dynamically tailors virtual content and messaging avatars in response to user behavior, with a focus on emotions and activities. Ongoing research efforts aim to enhance emotion detection accuracy using multimodal sources, such as voice, to ensure precise assessments. Additionally, the integration of biosensors and advanced models, such as NLTK, contributes to the nuanced spectrum of color changes, responsive to emotional intensity, even within states like happiness and mobile sensors for activity recognition. The incorporation of head-mounted display (HMD) technology amplifies user interaction, empowering users to interact with virtual flowers by initiating dynamic changes, including movement, message generation, and color alteration based on their interactions with virtual content.

ACKNOWLEDGMENT

This work was supported in part by the ITRC Support Program under Grant IITP-2023-RS-2022-00156354 and in part by the Metaverse Support Program to Nurture the Best Talents under Grant IITP-2023-RS-2023-00254529 funded by the Ministry of Science and ICT of Korea and the Institute of Information and Communications Technology Planning and Evaluation (IITP) and in part by the Ministry of Trade, Industry and Energy and Korea Institute for Advancement of Technology under Grant P0016038.

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